

Europe Beyond Aid: Evaluating Europe's Contribution to the Transfer of Technology and Knowledge to Developing Nations

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I. Objective

- Discuss European commitment to global technological development
 - Has its commitment fallen short?
 - What to do about it?
- Commitment gauged by
 - Public R&D (*to create development-relevant innovations*)
 - Intellectual Property Regulations (*to promote diffusion, or not impede it*)
 - Compliance with TRIPS Article 66.2 (*to create incentives for technology transfer to least developed countries by enterprises & institutions in Europe*)

Sound and Viable Technological Base

- Technology: important to productivity, standard of living, quality of life
- Ultimate aim: develop the knowledge base, innovative & absorptive **capacity** of developing nations – especially, the least developed.
 - Requires both internal and external efforts (Europe and other developed countries)
- Current state: indigenous **capacity** is lacking, lagging, or nascent.
- How to acquire that capacity?
 - Resources, inputs, access to knowledge
 - Improved productivity/efficiency in converting resources, inputs, and know-how into technological ‘outputs’
 - Forward-looking incentives: opportunity to make economic use of those outputs, whether commercially (domestic or foreign sales) or other purposes

Role of Europe (and public sector)

- Main source of technological **inputs** and **outputs** is the “North”
 - Some inputs/outputs generic
 - Others are specific to the *needs* of LDCs
- Obstacles
 - Market Size small (given purchasing power).
 - Private enterprises in the North have limited incentives to do South-relevant R&D
 - Public R&D for another country?
 - Using tax dollars to benefit foreigners
 - Tight budgets (aftermath of global financial crisis)
 - International free-riding
 - A classic public goods problem at the global level

International Resource Allocation Problem

- How to motivate governments to increase their commitment to global technological development?
 - And comply with international obligations?
- How to create incentives and opportunities for private enterprises and other institutions to transfer technologies to LDCs?
 - And foster greater Corporate Social Responsibility?
- How to provide favorable technology transfers to LDCs?
 - Affordability and access to *substantive* inputs & outputs
- What developing nations and least developed countries can do to attract technology transfers?
 - And strengthen their absorptive capacities, and utilize technologies effectively?



II. How do we assess European Commitment?

- Review **Article 66.2** Compliance
- Examine **Government Support for R&D**
- Assess impact of recent **IPR policies**

A. Technology Transfers to Least Developed Countries

- Article 66.2 Mandate (TRIPS)
- Since 2003, annual reports due on technology transfer (henceforth TT) activities, w/ full report every 3rd year.
- How seriously is this obligation taken?
- How seriously are the reports taken?

EU Guidelines:

“[T]he EU considers that relevant incentives can be identified as those that:

- Objective 1: Promote projects such as direct investment, licensing, franchising, sub-contracting, etc.
- Objective 2: Improve access to available techniques and industrial processes.
- Objective 3: Support joint research projects.
- Objective 4: Provide training in technology management and production methods.
- Objective 5: More indirectly, improve the absorption capacity of LDCs (capacity building).
- Objective 6: Encourage trade in technological goods.”

“Technology transfer refers here to the ways and means through which companies and organisations acquire technology from foreign sources. There are several types of technologies as well as several channels of transmission. Indeed, the acquisition by least developed countries (LDCs) of a sound and viable technological base does not depend solely on the provision of physical objects or equipment, but also on the acquisition of know-how, on management and production skills, on improved access to knowledge sources as well as on adaptation to local economic, social and cultural conditions.” (p. 1)

“Finally, it should be borne in mind that no technology transfer programme is specifically dedicated to least developed countries as such. EU initiatives are usually specific to countries/groups of countries/regions, since the EU strongly supports regional integration, which fosters better understanding and political and economic links between neighbouring countries.” (p. 2)

Table 4A. Developed Country Submission of Reports on Technology Transfer Activities (per Article 66.2 TRIPS)

Countries	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013*
Australia	Yes			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
New Zealand	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Norway	Yes		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Switzerland	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Canada	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Japan	Yes		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
US	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
European Union	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Austria	Yes				Yes	Yes	Yes	Yes	Yes	Yes	
Belgium					Yes	Yes	Yes	Yes	Yes		Yes
Czech Republic	Yes		Yes	Yes	Yes	Yes	Yes				
Hungary											
Denmark	Yes		Yes		Yes	Yes	Yes	Yes		Yes	
Finland	Yes	Yes	Yes		Yes	Yes	Yes	Yes	Yes	Yes	Yes
France		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Germany	Yes	Yes	Yes	Yes	Yes	Yes**	Yes	Yes	Yes		
Greece											
Ireland		Yes	Yes			Yes				Yes	Yes
Italy			Yes								
Luxembourg											
Netherlands	Yes		Yes								
Poland											
Portugal											
Slovakia			Yes	Yes					Yes	Yes	Yes
Spain	Yes	Yes	Yes		Yes	Yes	Yes	Yes		Yes	
Sweden	Yes	Yes	Yes		Yes	Yes	Yes	Yes	Yes	Yes	Yes
United Kingdom	Yes				Yes		Yes	Yes	Yes	Yes	Yes

* As of the time of this writing

** No narration; just a table of programmes submitted.

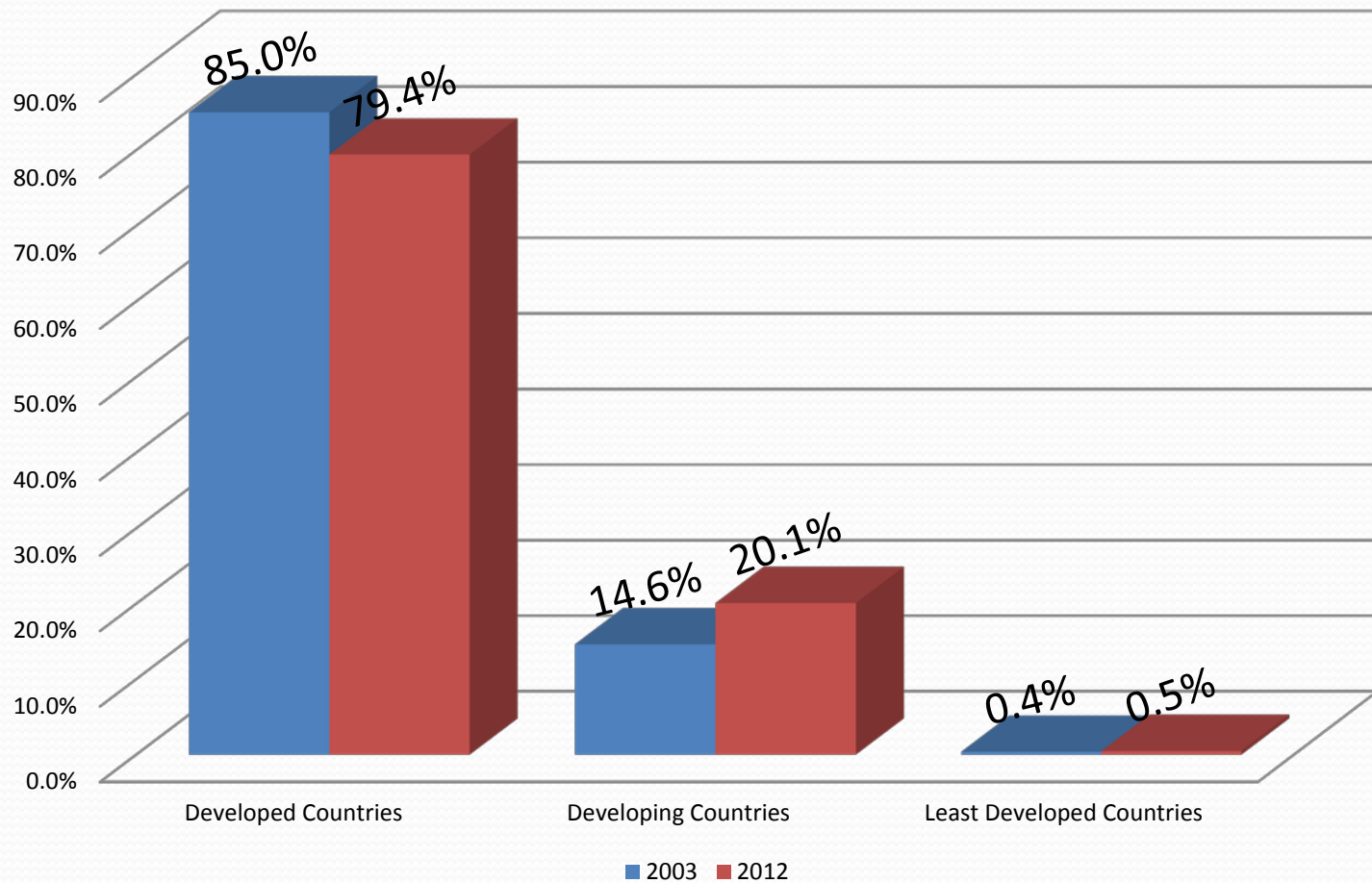
YES -- indicates that a report was submitted

Source: World Trade Organization (2003- 2014)

Criticisms

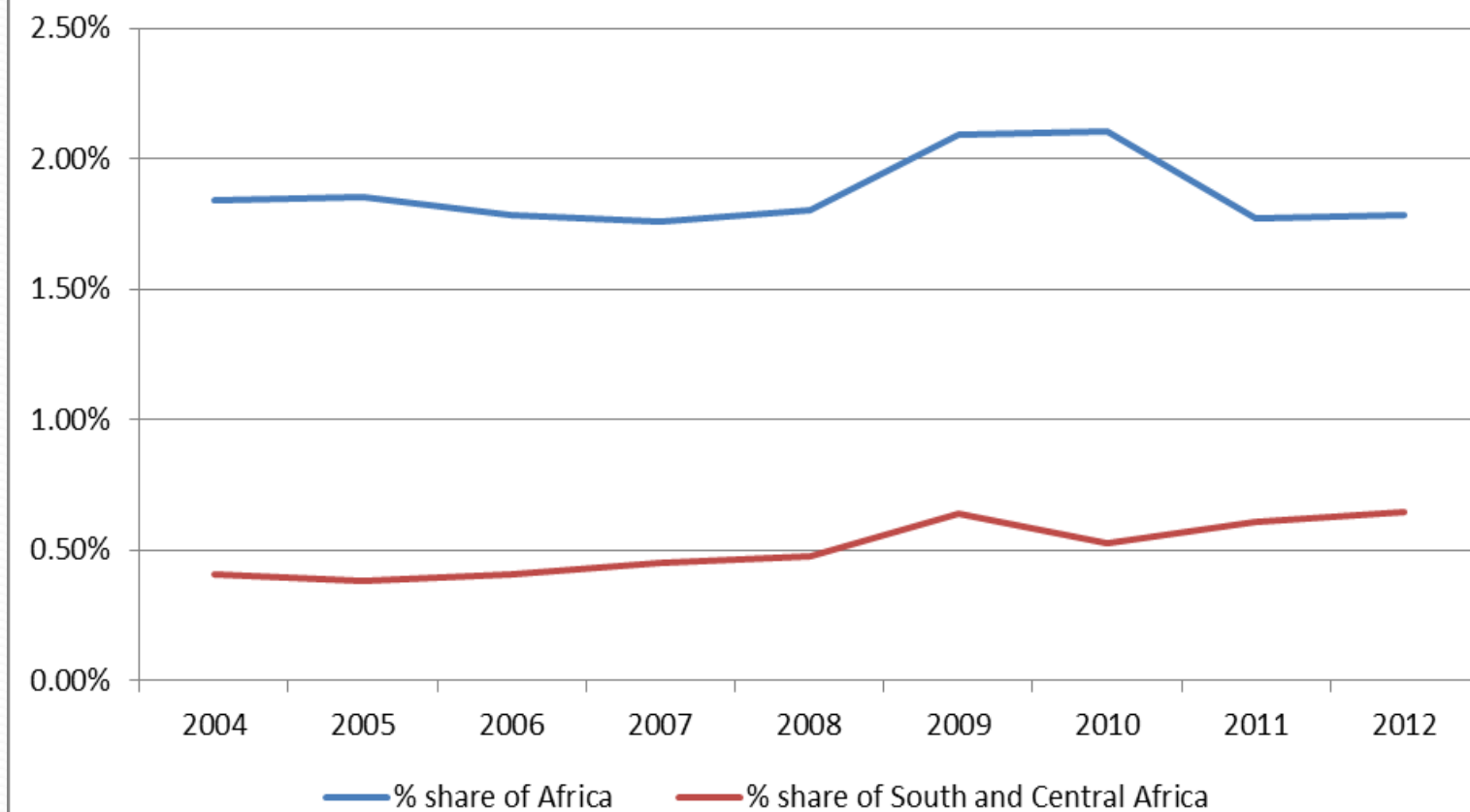
- Amounts are relatively small; programs not always substantive, or targeted to LDCs.
- Reports need more details and structure, and allow for an evaluation of TT programs or incentives.
- Much copying and pasting from previous years' reports.
- Evidence of increased TT is not apparent in recent data

**Europe 16 Exports of High-Tech Goods:
Percentage Distribution by Level of Economic Development of Destination**



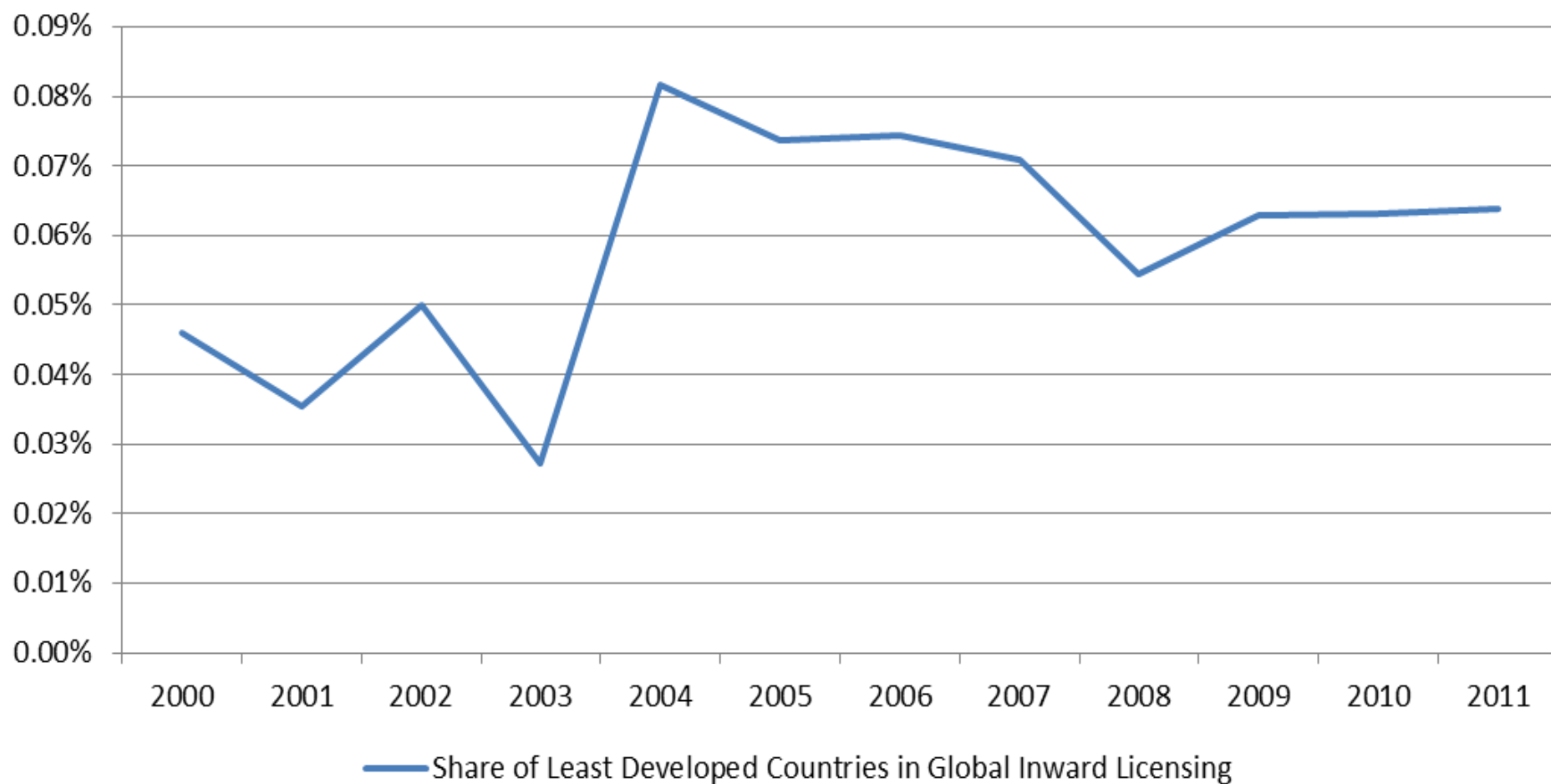
Source: UNCTAD Statistics

Share of EU27 Outward Stock of Foreign Direct Investment in Africa



Source: EuroStat

Percentage Share of Least Developed Countries in Global Inward Licensing



Source: UNCTAD Statistics

B. Government R&D support and IPRs

- CDI (Technology Component) measures:
 - Government R&D Support (2/3 weight)
 - Environment, Earth/Space Exploration & Exploitation, Advancement of Knowledge, Industrial Production & Tech., Energy, Health, Education, Culture/Political/Social Systems, Agriculture (discounted 25%), Defense (discounted 50%) – expenditures expressed as percentage of GDP
 - Subsidization of Business R&D as a % of GDP
 - Intellectual Property Rights (1/3 weight)
 - Non-patentability of software, plant & animals
 - Limitations on Patent Rights (compulsory licensing, opposition, exemptions for R&D, revocation)
 - Absences of IPR Extensions (e.g. TRIPS-plus FTAs, database protections, anti-circumvention of TPMs)

Country	Government R&D	IPRs	Overall score
South Korea	7.8	4.8	6.8
Denmark	7.8	4.2	6.6
France	7.5	4.6	6.6
Portugal	6.8	5.6	6.4
Japan	6.7	5.4	6.2
Norway	6	5.2	5.7
Finland	6.1	4.7	5.7
Austria	5.9	5.1	5.6
Spain	5.4	5.5	5.4
Czech Republic	5.7	4.8	5.4
Canada	4.5	6.9	5.3
Netherlands	5.6	4.3	5.2
The Rest (incl. South Korea)	5.4	4.7	5.2
Germany	5.5	4.2	5.1
Europe 16	5.3	4.4	5
Switzerland	4.8	5	4.9
Europe 21	5.1	4.5	4.9
United States	5	4.1	4.7
Australia	3.7	6.7	4.7
Sweden	4.8	3.9	4.5
Belgium	4.3	4.7	4.4
New Zealand	3.1	7	4.4
United Kingdom	4.3	4	4.2
Luxembourg	4	4.4	4.1
Italy	3.7	4.4	3.9
Ireland	3.6	4	3.8
Hungary	2.5	4.7	3.2
Greece	1.8	4.6	2.7
Slovakia	1.6	4.7	2.6
Poland	1.6	4.5	2.5

Europe's Performance compared to other Developed Countries

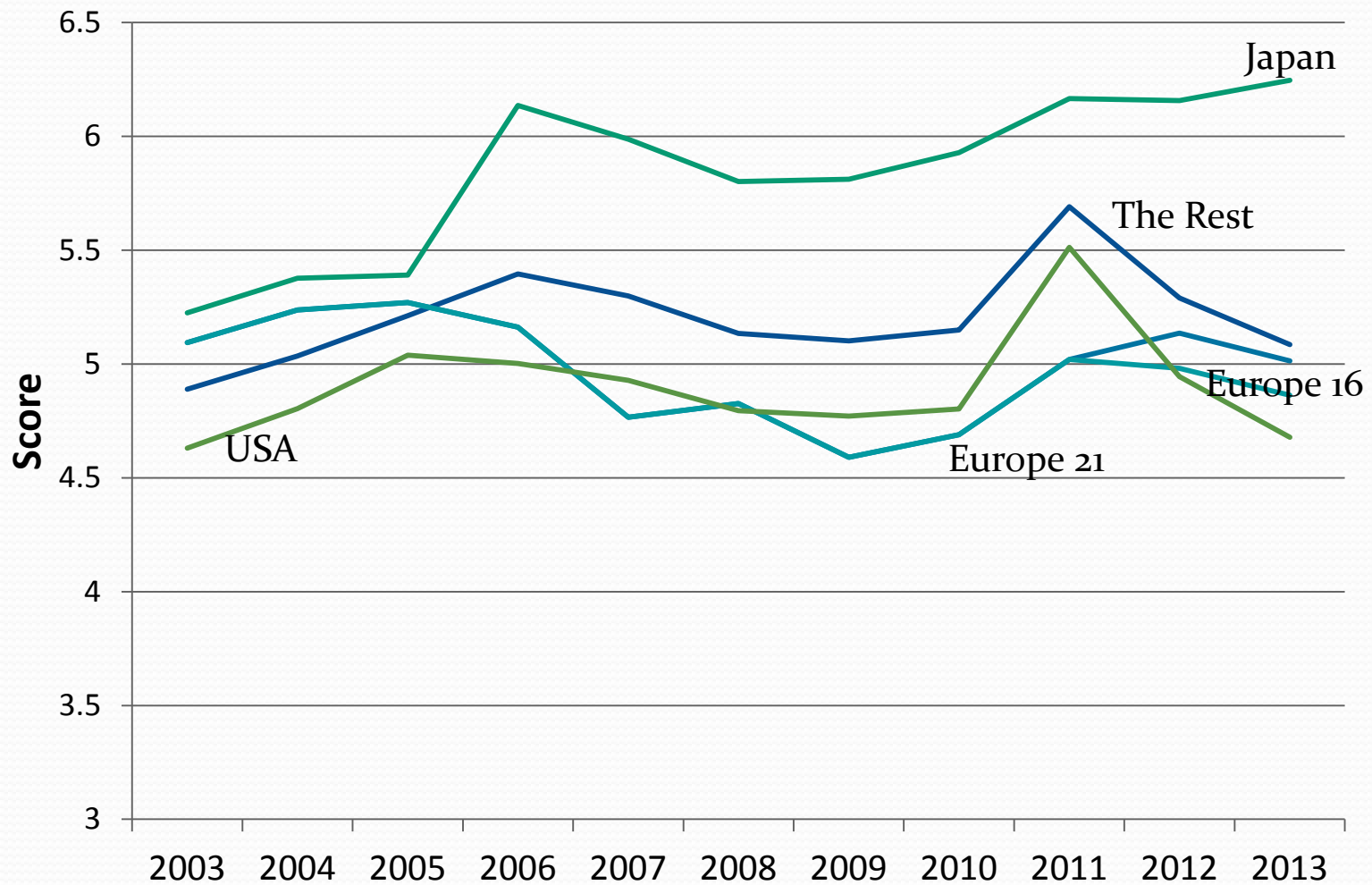
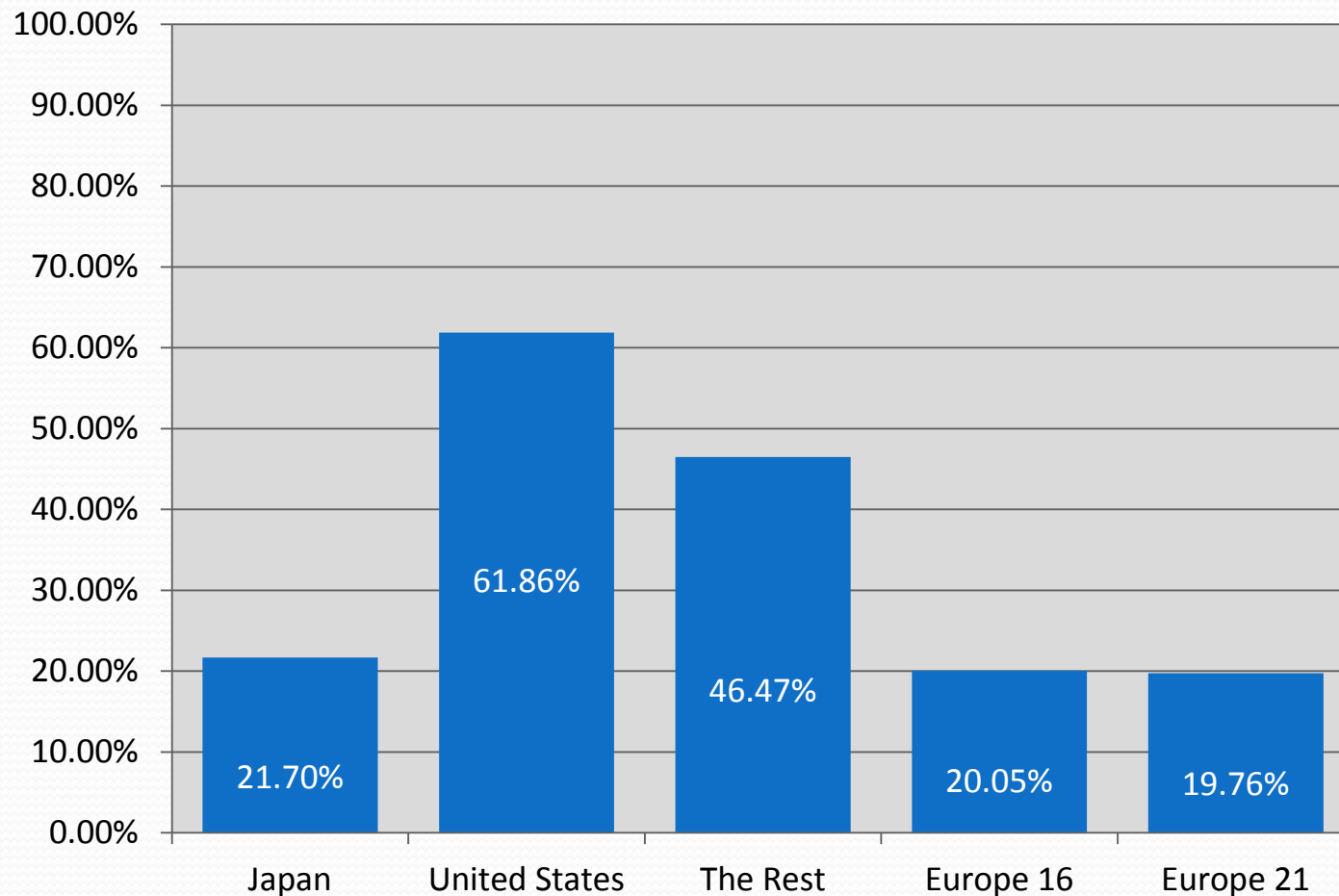


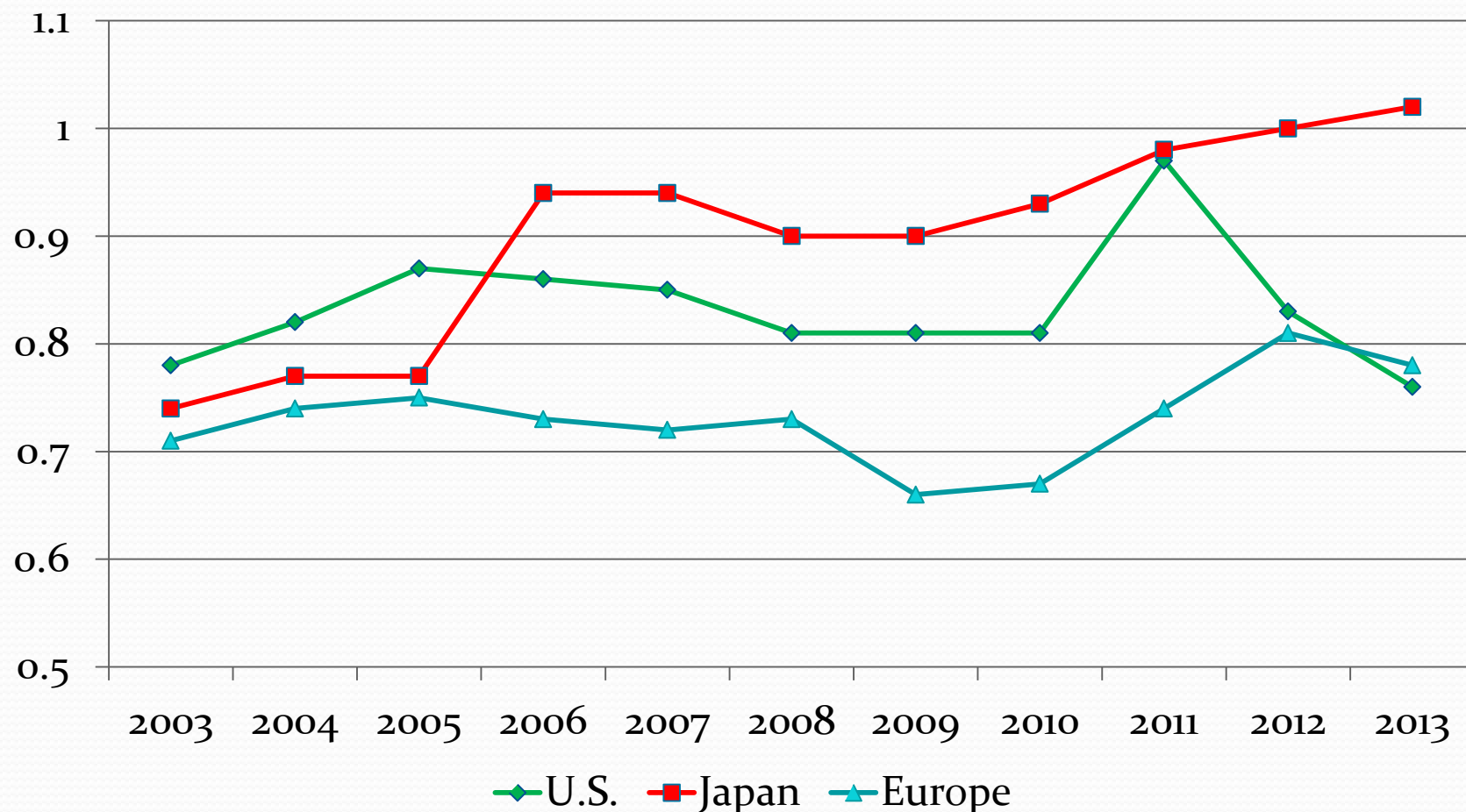
Table 1B. Technology Component, Government R&D Sub-Score											
Government R&D	2013	2012	2011	2010	2009	2008	2007	2006	2005	2004	2003
Japan³	6.7	6.5	6.4	6	5.9	5.8	6.1	6.1	5	5	4.8
South Korea	7.8	8.1	7.3	6.9	5.8	6.4					
United States³	5	5.4	6.3	5.3	5.3	5.3	5.5	5.6	5.7	5.3	5.1
Europe 16¹	5.3	5.5	5.4	4.9	4.7	5	4.9	5	5	5	4.7
The Rest³	5.3	5.6	6.2	5.4	5.3	5.4	5.6	5.7	5.4	5.2	4.9
Europe 21²	5.1	5.3									
The Rest (inc. Korea)	5.4	5.7	6.2	5.5	5.3	5.4					
Median	4.8	5.1	5.4	4.8	4.4	4.8	4.8	4.8	4.9	4.9	4.5

Table 1C. Technology Component, Intellectual Property Rights sub-score											
IPRs	2013	2012	2011	2010	2009	2008	2007	2006	2005	2004	2003
Japan³	5.4	5.4	5.7	5.7	5.7	5.7	5.7	6.1	6.1	6.1	6.1
South Korea	4.8	4.8	5.5	5.5	7	7					
United States³	4.1	4.1	3.9	3.9	3.8	3.8	3.8	3.8	3.8	3.8	3.8
Europe 16¹	4.4	4.4	4.3	4.3	4.4	4.5	4.5	5.5	5.8	5.8	5.8
The Rest³	4.7	4.7	4.7	4.7	4.6	4.6	4.6	4.8	4.8	4.8	4.8
Europe 21²	4.5	4.4									
The Rest (inc. Korea)	4.7	4.7	4.7	4.7	4.7	4.7					
Median	4.7	4.7	4.7	4.7	4.7	4.8	4.8	5.9	6.1	6.1	6.1

Public Sector R&D: Development-Friendly Spending as a % of Total (non-defense) Government R&D



Total Government R&D Support as a percentage of GDP



Source: CDI 2013

C. IPR Developments & Concerns

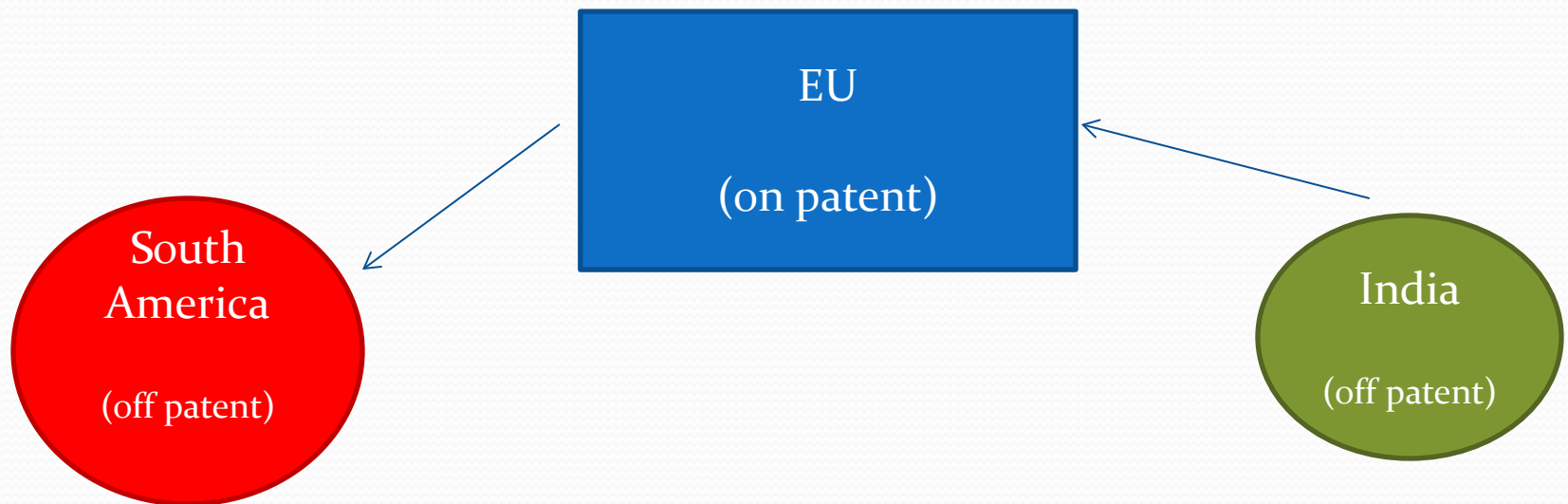
- CDI Index:
 - Patenting of Software, Plant & Animals
 - Directive 98/44/EC, Article 52(2) of EPC
 - Lack of Limitations on Patent Rights: Compulsory Licensing, Revocation for Non-working, Opposition, Research Exemptions
 - IPR Extensions: Database Protection, Anti-circumvention laws, and TRIPS-plus provisions
- These components in CDI refer to Europe's IP regime, not that of developing countries:
 - Purpose: Assess how European IPRs facilitate or impede technology diffusion to the 'South'

TRIPS-plus provisions

- EU Free Trade Agreements (FTAs) with developing country partners contain provisions ‘beyond TRIPS’
- TRIPS-plus may push Southern IPRs above what is optimal at their stage of economic development
 - E.g. restrict compulsory licensing, parallel importation, or expand patent subject matter
 - Lengthen Test Data Exclusivity (delay entry of generics or ‘biosimilars’); allow new therapeutic uses

Enforcement

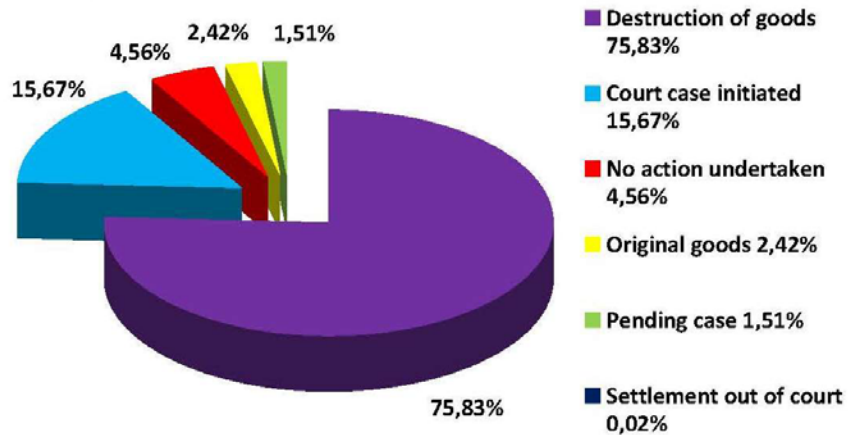
- EC Council Regulation 1383/2003: customs actions
- Prohibits exports, or re-exports, from EU of IP-infringing goods, even for in-transit goods



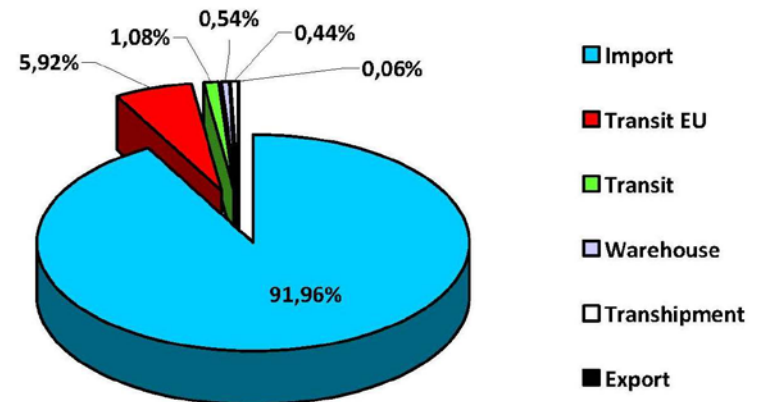
Enforcement

From EC (2010) *Report on EU Customs Enforcement* ...

Breakdown of results by cases

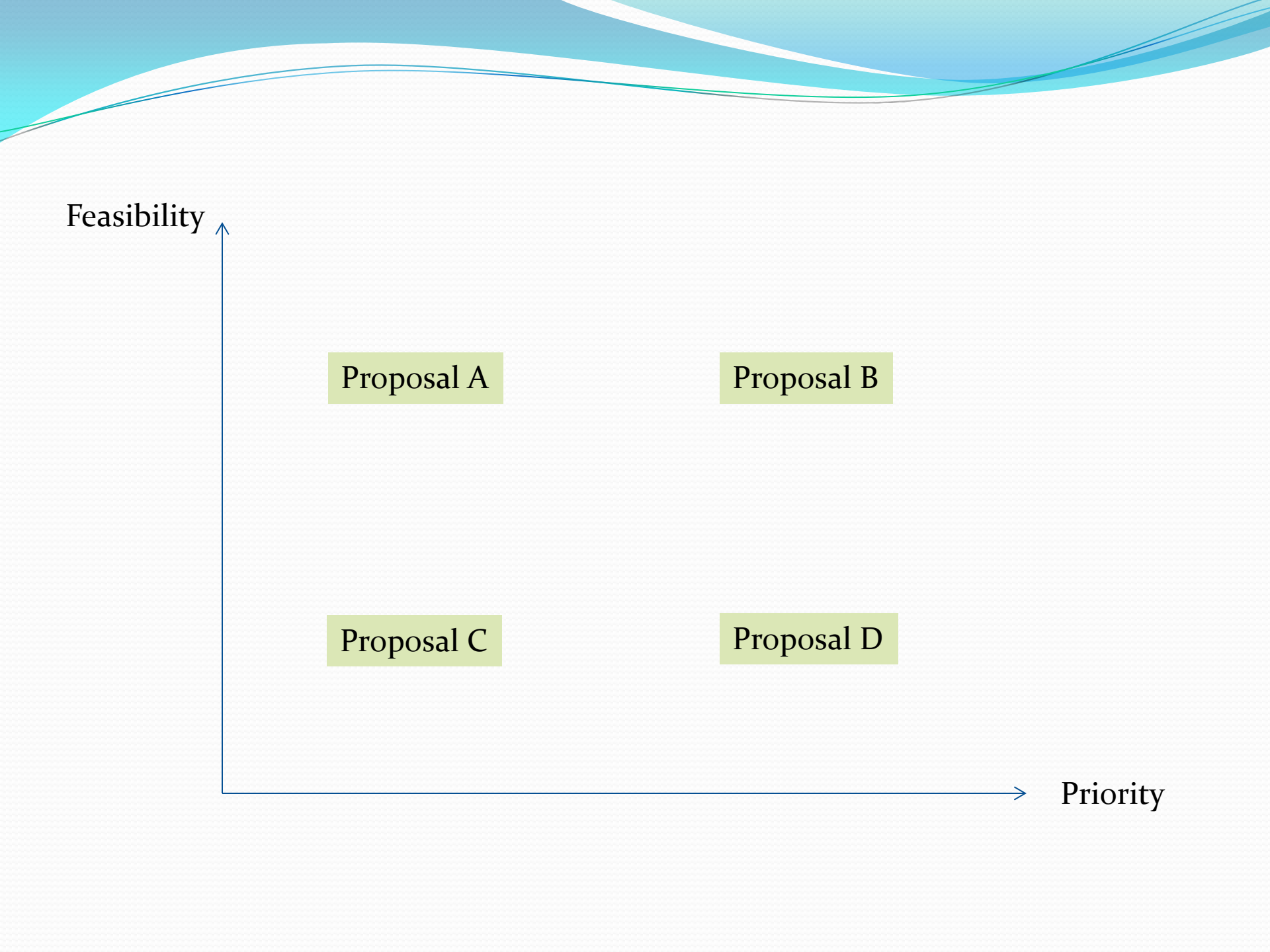


Breakdown of cases by procedure



III. Policy Proposals

- Main take away points:
 - European commitment to technological development in the developing world, especially in LDCs, demonstrates shortcomings:
 - *Development-friendly public R&D spending is relatively low*
 - *Tech transfers to developing world impeded due to stringent IPRs.*
 - *Compliance with Article 66.2 is ambiguous.*
- What can be done to raise it?
- Which policies are feasible and high priority?
 - Degree of Difficulty and Degree of Importance
(cf. Framework in Atlantic Council *Transatlantic Trade and Investment Partnership*)
- Responsible jurisdictions:
 - European Union (EU) or National member governments?



Feasibility

Proposal A

Proposal B

Proposal C

Proposal D

Priority

Policy Proposal	Feasible (1 – 5)?	Priority (1 – 5)?
Expand Public R&D in Development-friendly areas		
Scale back on stringent IPRs & harmonization (specify)		
Reduce Test Data Exclusivity periods		
Exempt in-transit goods from Regulation 1383		
Conduct Development Impact Studies prior to FTAs		
Create ‘Instructions’ and ‘Targets’ for Article 66.2 obligations		
Provide substantive Tech Transfers and Incentives to LDCs		
Establish an official Tech Transfer Office in the EU		
Make an Affirmative Declaration on Technology Transfer		
Tailor R&D projects and IPR policies to local circumstances		
Reduce piracy and infringement of IPRs		
Promote copyright flexibilities and exemptions in LDCs		

5 = High priority or Highly Feasible
1 = Low Priority or Quite Difficult to implement

Policy Proposal	Feasible (1 – 5)?	Priority (1 – 5)?
Use compulsory licensing for ‘export’ (Regulation 816/2006)		
Offer risk insurance and credit to firms’ TT in LDCs		
Use patent buyouts or advanced marketing commitments		
Relax anti-circumvention of Technology Protection Measures		
More support for open innovation in developing nations		
Other Recommendations ...		

5 = High priority or Highly Feasible
 1 = Low Priority or Quite Difficult to implement